

Mark Tangchaisuk

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EDUCATION

Penn State University

Pennsylvania

Bachelor of Science, Computer Science (Senior) | Minor: Engineering Leadership Development Aug 2023 - Expected May 2027

- GPA: 3.82 | Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026
- Relevant Coursework: Operating Systems (CMPSC 473), Theory of Computation (CMPSC 464), Computer Vision (CMPEN 454), Design & Analysis of Algorithms (CMPSC 465), Numerical Computations (CMPSC 451), Systems Programming (CMPSC 311), Data Structures & Algorithms (CMPSC 360), Digital Design (CMPEN 270)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C, C++, SQL, Dart, HTML

Tools & Platforms: React, PixiJS, Node.js, Colyseus, Azure, Databricks, PySpark, Delta Lake, PyTorch, Roboflow, ROS, Vitest, Playwright, Flutter, React Native, Expo, Git, Linux, VS Code

EXPERIENCE

Accenture

Bangkok, Thailand

Business Analyst and Data Engineer Intern

June 2026 - July 2026

- Designed and built components of a new Modern Data Platform (MDP) for Kasikornbank (KBANK), one of Thailand's largest banks, using Azure Databricks and SQL to consolidate core banking data into a layered lakehouse.
- Developed ingestion pipelines in PySpark that loaded source tables from MSSQL into Delta Lake, and implemented transformation logic covering deduplication, type casting, and schema validation.
- Authored a bilingual English and Thai data dictionary for payment and transaction tables, writing field definitions clear enough for both business and engineering readers.
- Skills: Databricks, SQL, PySpark, Delta Lake, Azure, Data Modeling, Technical Documentation, Stakeholder Communication

Business Applications Co., Ltd.

Bangkok, Thailand

Data Scientist Intern

May 2025 - Aug 2025

- Built data pipelines and delivered business insights using Tableau, Power BI, Azure, and Databricks, presenting technical findings to executives in plain language.
- Developed machine learning models in PyTorch and applied them to real world datasets for predictive analysis and reporting.
- Skills: Tableau, Power BI, Azure, Databricks, PyTorch, Data Pipelines, Machine Learning, Technical Presentation

Advanced Vehicle Team, Penn State University

University Park, PA

Perception Team Member

Aug 2025 - Present

- Developed perception systems for an autonomous vehicle using ROS, building and training computer vision models in Roboflow for depth estimation and object detection under an NDA-protected research project.
- Skills: ROS, Roboflow, Computer Vision, Machine Learning, Python, Robotics Integration

HUB Retail Dining, Burger King, Penn State University

University Park, PA

Crew Member

Sept 2025 - May 2026

- Trained new employees on food safety standards, register operation, and store procedures. Served hundreds of students daily while balancing a full course load.

PROJECTS

Top Charge, Real-Time Multiplayer Strategy Fighter

May 2026 - Present

TypeScript, React, PixiJS, Node.js, Colyseus, Vitest, Playwright | playtopcharge.com

- Built a browser based simultaneous reveal fighting game with a resource economy, upgrades, and a shop, playable in normal, quick, and vs AI modes. Created all game art, logic, and hosting.
- Architected an authoritative Colyseus and Node server with deterministic game logic shared between client and server; clients send only intents, server validates every payload and broadcasts state via synchronized schemas.
- Built the client in React and PixiJS with procedural animation, particle effects, hit stop, and camera shake, plus an internationalized UI in English and Japanese. Deployed to Oracle Cloud behind Nginx and TLS. Wrote a 110+ test Vitest suite and a Playwright end to end harness.

Alure, AI Styling Platform for College Students

March 2026 - Present

React Native, Expo, JavaScript, Google Gemini API | github.com/Markboon123/Alure

- Built a mobile app that generates outfit recommendations from a user's wardrobe using the Google Gemini API for outfit generation and a chat based styling assistant, with a feedback loop so recommendations improve over time.
- Collaborated with a five person team through GitHub, taking the product from wireframes to a working build on iOS and Android.